

OBSTETRICS

Is obstetric and neonatal outcome worse in fetuses who fail to reach their own growth potential?

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ABSTRACT

Objectives To determine the perinatal outcome of fetuses who had birthweights less than that expected from early third trimester ultrasound scanning.

Design Retrospective estimation of centile fetal weight at early third trimester ultrasound scanning compared with actual centile birthweight corrected for gestational age, parity and sex.

Setting Teaching Hospital Obstetric Unit, London.

Subjects 197 unselected women with singleton cephalic pregnancies who were delivered at term in our unit between October 1989 and May 1990.

Main outcome measures CTG abnormality, need for fetal blood sampling in labour, meconium-staining of the amniotic fluid, mode of delivery, Apgar scores at 1 and 5 min, need for transfer of baby to neonatal unit, and need for neonatal intubation of the neonate at delivery.

Results An actual birthweight >5% less than the birthweight estimated from ultrasound scanning identified 44 babies (22%) with an increased risk of CTG abnormalities ($\chi^2 = 8.38$, $P < 0.0025$; Odds ratio (OR) = 2.54; 95% CI 1.36 to 4.78) and need for operative delivery ($\chi^2 = 5.81$, $P < 0.0125$; OR = 1.94; 95% CI 1.15 to 3.27), when compared with the remainder of the sample. Overall 14 (32%) of this group had birthweights above the 50th centile. A group of 44 babies selected as being the smallest for gestational age, without reference to growth pattern, had a similar excess morbidity. (All this group had birthweights below the 39th centile.)

Conclusions This study supports the hypothesis that *in-utero* fetal growth pattern is as important for perinatal outcome as being small for gestational age *per se*.

Most current definitions of intrauterine growth retardation (IUGR) involve a categorization of birthweight for gestational age. Thus babies <10th centile birthweight for gestational age, are commonly referred to as 'growth retarded'. However 'small-for-gestational age' (SGA) is a better term for this type of categorization as it is likely that many babies <10th centile are small but normal (Goodlin 1990).

It has been suggested that the term IUGR should be reserved for those fetuses that do not reach their full growth potential, even if they are above the 10th centile for weight at birth. It has been further suggested that babies of normal weight who were genetically programmed to be larger but have experienced growth restraint because of *in-utero* factors, are more at risk than the SGA infant *per se* (Altman & Hytten 1989; Campbell 1989; Pearce & Campbell 1985; Daikoku *et al.* 1979; Hull *et al.* 1978). However, apart from anecdotal case reports (Bradley *et al.* 1988), we could find no record of attempts to analyse perinatal morbidity by growth pattern, rather than by centile birthweight. We therefore carried out a retrospective survey to test the hypothesis that perinatal performance was related to growth potential as determined by an early third trimester ultrasound scan.

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Subjects and methods

We studied 197 unselected women with a singleton pregnancy and a cephalic presentation at delivery, who had a routine early third trimester ultrasound scan (between 28 and 34 weeks gestation as part of an existing policy to detect intrauterine growth retardation), and who were delivered at term at our hospital between October 1989 and May 1990. All women had an ultrasound scan at 18–20 weeks gestation to confirm dates and to screen for major structural anomalies. (All scans were performed at a hospital antenatal clinic on a Siemens SL2 machine.) Measurements of fetal biparietal diameter (BPD), head circumference (HC), abdominal circumference (AC) and femur length (FL) were made at the early third trimester scan. From these measurements a fetal weight was estimated (Hadlock *et al.* 1985) and the standard deviation and centile weight for the gestational age at the time of the scan obtained from published data (Jeanty *et al.* 1984). An estimate of birthweight was then calculated (to the nearest 50 g) by assuming no change in centile between scan and birth and then using the nomogram described by Altman & Coles (1980), corrected for gestational age, sex and parity. Actual birthweight (ABW) and estimated birthweight (EBW) were then compared, and the percentage change between the actual and the estimated birth-

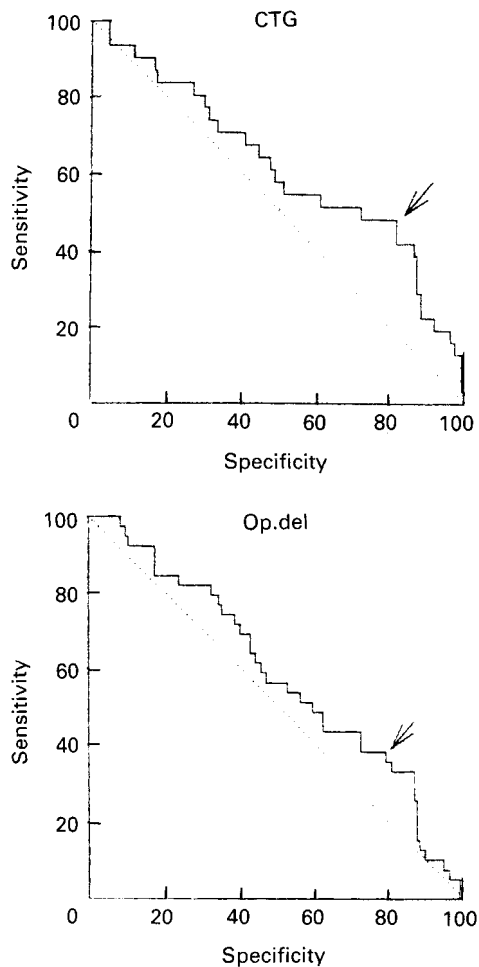


Fig. 1. Receiver operator curves for cardiotocograph (CTG) abnormality and need for operative delivery using percentage change between estimated birthweight and actual birthweight as screening test. (For explanation of arrow, see text).

weights was calculated. (Percentage change = $(ABW-EBW)/EBW \times 100$).

The presence of an abnormal cardiotocograph pattern (CTG) in labour, meconium staining of the amniotic fluid, intrapartum fetal blood sampling and pH (when taken), the mode of delivery, Apgar scores at 1 and 5 min, the need for neonatal intubation and admission to the neonatal intensive care unit (NICU) were noted. This information was obtained from the St Mary's Maternity Information System.

Discriminant analysis was performed using commercially available statistical software (SPSS PC+; SPSS Inc., Chicago, USA), comparing the percentage change as calculated above, and the centile birthweight corrected for sex, parity and gestational age. Receiver-operator curves were constructed for percentage change for each outcome parameter.

Results

The actual birthweight (ABW) was greater than the estimated birthweight (EBW) in 113 babies (57%), and was lower in 85 (43%), ($P > 0.05$; ns). The mean ABW was 3384 g (SD 487 g) and the mean EBW was 3308 g (SD 332 g), ($t = 1.83$; ns).

There were only 10 babies (5%) born with birthweights below the 10th centile as defined by Altman & Coles (1980). There were no perinatal or neonatal deaths in this series and none of the babies had hypoxic-ischaemic encephalopathy (HIE).

Discriminant analysis showed that percentage change between ABW and EBW was significantly related to CTG abnormality ($P = 0.02$), and need for operative delivery ($P = 0.05$). Centile birthweight was significantly related to abnormal CTG ($P = 0.01$) only. No other significant correlations were found using discriminant analysis.

Using receiver-operator curves (Fig. 1), in which sensitivity was plotted against specificity for all 196 possible cut-offs of percentage change, showed that a drop of more than 5% from the EBW to ABW (arrowed in figure) gave a sensitivity of 48% and specificity of 82% for CTG abnormality, and a sensitivity of 36% and specificity of 80% for operative delivery. This cut-off selected 44 babies (22%), and in this group there was significantly more CTG abnormality ($\chi^2 = 8.38$, $P < 0.0025$; odds ratio (OR) = 2.54; 95% CI 1.36 to 4.78), and need for operative delivery ($\chi^2 = 5.81$, $P < 0.0125$; OR = 1.94; 95% CI 1.15 to 3.27) than in the rest of the sample. When the 44 babies with the lowest birthweight for gestational age were studied, they were all below the 39th centile as defined by Altman & Coles (1980). They also had significantly more abnormal CTGs ($\chi^2 = 6.83$, $P < 0.005$; OR = 2.36; 95% CI 1.25 to 4.7) than the rest of the sample but were not more likely to need operative delivery. When the two groups were compared, there was a significant difference between their mean centile birthweights ($t = 3.57$; $P < 0.001$). (see Fig. 2). Overall 14 babies (32%) in the group with a $>5\%$ fall had actual birthweights above the 50th centile (25 babies fell into both groups).

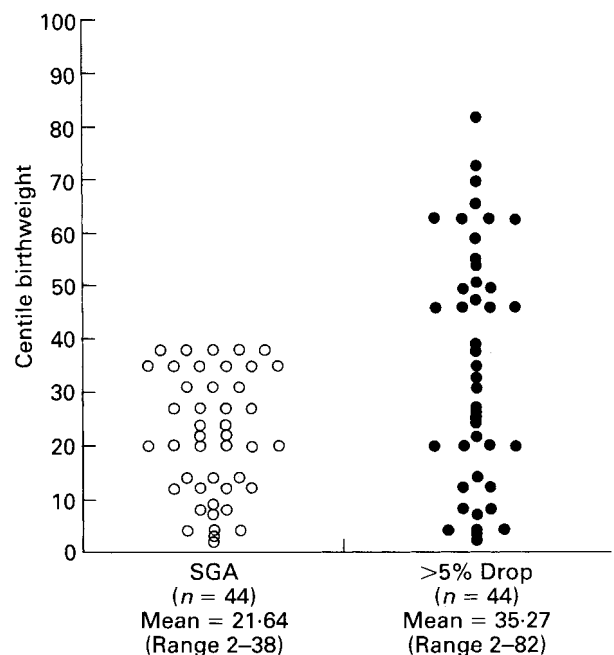


Fig. 2. Comparison of centile birthweights ($t = 3.57$; $P < 0.001$). Small-for gestational age (SGA) = lowest 22% of sample by centile birthweight. $>5\%$ Drop = babies with actual birthweight $>5\%$ less than estimated birthweight.

Discussion

The results of this study support the hypothesis that growth pattern is as closely related to perinatal performance as actual size at delivery. There were significant increases in CTG abnormalities and operative delivery in the 22% of our population with a >5% fall in centile weight during the third trimester, even though 32% of them were above the 50th centile (weight for gestational age) at birth. This latter group of babies would not normally be categorized as 'at risk' by most current clinical criteria. Those babies who have a fall in growth rate as defined here have the same intrapartum and neonatal morbidity as those which are SGA.

We now need to examine the hypothesis that the fall in growth rate can be detected antenatally by ultrasound, as opposed to determined after birth. We are currently conducting a prospective trial to test this hypothesis. Two third trimester ultrasound scans will be performed in an attempt to diagnose any fall in growth rate and to ascertain whether an ultrasound detected fall will be predictive of the abnormal outcomes identified in this retrospective study. Doppler studies will also be performed at the time of the second scan. Even if a fall in growth rate can be measured reliably, any benefit in outcome will depend upon effective management strategies. Whether such strategies are available is at present open to question. This issue will be examined by a subsequent trial in which 50% of the scan results are revealed and 50% concealed.

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